

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Semester Exam–2024

Course: B. Tech.

Semester :V

**Branch : Electronics and Telecommunication Engineering/ Electronics and
Communication Engineering**

Subject Code & Name: BTETPE504A Analog Circuits

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
 - 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
 - 3. Use of non-programmable scientific calculators is allowed.*
 - 4. Assume suitable data wherever necessary and mention it clearly.*
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	CO	Marks
Q.1 Solve Any Two of the following.		12
A) Discuss Single Slop Analog to Digital Converter?	CO1	6
B) Explain the Non-Inverting Comparator?	CO6	6
C) Write short note on Square wave generator?	CO3	6
Q.2 Solve Any Two of the following.		12
A) Discuss the factors affecting stability of oscillators?	CO6	6
B) Write a short note on Integrator and derive its expression?	CO3	6
C) Write a short note on Averaging Amplifier and derive its expression?	CO6	6
Q.3 Solve Any Two of the following.		12
A) Explain Positive Precision full Wave Rectifier?	CO2	6
B) Write a note on: Voltage to current converter using op-amp with Ground Load.	CO2	6
C) Explain Inverting zero crossing detector?	CO2	6
Q.4 Solve Any Two of the following.		12
A) Derive an expression for Inverting amplifier?	CO5	6
B) What is an operational amplifier? Explain the Operational Amplifier with the help of its block diagram?	CO5	6
C) Classify different types of oscillator?	CO8	6

Q. 5	Solve Any Two of the following.		12
A)	Discuss frequency response curve of amplifier?	CO2	6
B)	Explain the different types of feedback in Operational Amplifier?	CO7	6
C)	Explain R-2R Resistive Ladder Network?	CO8	6

***** End *****